Highlights

The environmental-contour benchmark re-decided exactly: 150 submitted contours as exact polygons, 1.8 million hourly sea states classified in exact arithmetic, and the counts the organizers printed[⋆]

Carlos Toledo

- Every submitted contour read as the exact polygon its decimal digits denote.
- 1.8 million hourly sea states classified inside, outside or on, every sign decided.
- 173 of 176 printed counts reproduced to the integer; 2 traced to a rewritten file.
- 30 of 150 polygons cross themselves; 49 are closed by an edge nobody listed.
- The benchmark's expected numbers certified as enclosures; its 20 and 1 are exact.

The environmental-contour benchmark re-decided exactly: 150 submitted contours as exact polygons, 1.8 million hourly sea states classified in exact arithmetic, and the counts the organizers printed

Carlos Toledo[a,*]

[a] *Independent researcher, the cert-machine project,*

Abstract

The benchmarking exercise for environmental contours of Haselsteiner et al. (2021) asked nine groups to submit 1-, 20- and 50-year contours for six hourly metocean datasets and scored each submission by counting the observations that fall outside it, with a floating-point point-in-polygon test. This paper re-decides that count. Each of the 150 submitted contour files is read as the polygon its decimal digits denote; each of the 1,840,824 hourly sea states is classified inside, outside or on that polygon under the even-odd rule the organizers' script used, every sign decided — in floating point where a proved forward-error bound fixes it, in integer arithmetic otherwise — and the organizers' counting procedure is then applied as their script defines it. Of the 176 numbers printed in the benchmark's two tables, 173 are the exact count to the integer; one more is the exact mean once two observations lying exactly on a contour are counted outside, which is what the float test does; two, the one-year row of one contribution, are not the count of any file the repository now holds, and the repository's own history traces them to three files rewritten after the count was made. 30 of the 150 polygons cross themselves, up to 26 times, and 49 are not closed in the file, so the region that was counted includes an edge nobody listed; no observation changes side between the even-odd and the nonzero-winding rule, so the printed

*Draft for review, version 0.1 of 2026-10-01. The author list and the CRediT roles are to be agreed. Every number in this manuscript is read at build time from the records named in the Data availability statement (repository commit 9349d8bad505); the prose is authored and should be reviewed like any other.

*Corresponding author.

Email address: carlos@carlostoledo.co (Carlos Toledo)

counts do not depend on that choice. The benchmark's expected numbers are certified: its 197, 11.5, 492 and 12 are the roundings of enclosures, and its 20 and 1 are exact. Nothing is claimed about which contour construction is better. The contribution is the decision grammar — a verdict with its witness, and a file's defects reported rather than counted over — and a public record that any reader can re-run.

Keywords: environmental contours, metocean design, benchmarking, point in polygon, exact geometric computation, verified computation, IFORM

---

1. Introduction

An environmental contour is a closed curve in the plane of two metocean variables — significant wave height against zero-up-crossing period, or wind speed against wave height — along which a structure is analysed for the sea states of a given return period (Winterstein et al., 1993; DNV, 2021; ISO, 2015). Several constructions compete: the inverse first-order reliability method (Winterstein et al., 1993), inverse second-order variants (Chai and Leira, 2018), direct sampling (Huseby et al., 2013) and highest-density regions (Haselsteiner et al., 2017), built on joint models whose fitting is itself a field (Jonathan and Ewans, 2013; Ross et al., 2020), and differing in what probability the contour encloses (Mackay and Haselsteiner, 2021). To compare them on common ground, Haselsteiner et al. (2021) organised a benchmarking exercise: six hourly datasets, three from buoys and three from a hindcast, were distributed with ten or twenty-five years each, a further ten or twenty-five years were retained, and nine groups submitted their 1-, 20- and 50-year contours as lists of vertices. The organizers then counted, for every contour, the observations of the provided and retained years together that fall outside it, printed the count beside the number expected for the contour's definition, and left the comparison of methods to the reader.

That count is the benchmark's metric, and it is a finite, decidable quantity: given a polygon, a rule for "inside", and a list of points, the number outside is an integer. Whether a program returns that integer depends on three things which the exercise, like any exercise of its kind, left to its software: whether the file denotes a polygon at all, which rule for "inside" is applied and what it answers on the boundary, and whether the arithmetic that evaluates the rule gets every sign right. The organizers' procedure, which is public, fixes the rule by the library it calls — the crossing-number test of matplotlib's `Path.contains_points` (Hunter, 2007) — and leaves the arithmetic to that library's double precision. Floating-point orientation

tests are known to answer wrongly near degeneracy, and to do so silently (Kettner et al., 2008; Shewchuk, 1997); the remedy, exact geometric computation behind a floating-point filter (Yap, 1997; Fortune and Van Wyk, 1996), is standard in computational geometry and, to our knowledge, has not been applied to a metocean benchmark.

This paper applies it. Every submitted file is read as the polygon its decimal digits denote, exactly; every hourly sea state is classified inside, outside or *on* that polygon with every sign decided; the organizers' counting procedure is reproduced as their script defines it; and each number they printed receives a verdict with the quantity that justifies it. Three things motivate the effort beyond the metric itself. First, a floating-point test has no third answer: a point on an edge is returned on one side or the other by rounding, and in this benchmark two observations do lie exactly on a submitted contour. Second, the exact reading surfaces properties of the files that a float count passes over — polylines that cross themselves, polygons closed by an edge nobody listed, vertices repeated, a file rewritten after it was counted — which bear on what the printed number is a count of. Third, a benchmark is the one place where a number ought to be a decision rather than an observation, because it is the number on which methods are compared.

The contribution is therefore not a new contour method and not a judgement on any existing one. It is (i) a reading of each submitted file as an exact polygon with its diagnostics, (ii) an exact classification of every observation with *on* as a third answer, (iii) the organizers' procedure reproduced and every printed number decided against it, (iv) the benchmark's expected numbers certified, (v) numbers the benchmark did not print — the out-of-sample counts on the retained years alone, and the highest wave along each contour against the highest observed — and (vi) the public, re-runnable record of all of this. The word *certified* is used in this paper in its mathematical sense only, for a statement with a machine-checkable proof: an enclosure computed with outward rounding, or an integer decided with every sign certain. A count obtained that way is said to be *decided*, and a printed number read against it *re-decided*. Section 2 describes the exercise and its procedure; Section 3 the exact re-decision; Section 4 the results; Section 5 what a certificate adds to a benchmark of this kind and what a next exercise could require.

Table 1: The 11 contour sets of Exercise 1, as the benchmark's Table 2 lists them: the construction (IFORM and ISORM, the inverse first- and second-order reliability methods; DSCM, direct sampling; HDCM, highest-density contours; IDSCM and DIFORM, the inverse directional simulation and direct-IFORM variants of contributions 3 and 5), the joint model as the benchmark abbreviates it, the organizers' class, the number of files, and the range of listed vertices per file.

set	authors	construction	model	class
1	W. Chai, B. Leira	ISORM	GHM	total
2	G. Clarindo, C. Guedes Soares	DSCM	GHM	marginal
3	Á. Hannesdóttir, N. Dimitrov	IDSCM	GHM	total
4	A. F. Haselsteiner, A. Sander, J.-H. Ohlendorf, K.-D. Thoben	HDCM	GHM	total
5	G. de Hauteclocque	DIFORM	PPOTM	marginal
6	E. Mackay, P. Jonathan	IFORM	SRM	marginal
7	C. Qiao, A. Myers	IFORM	GHM	marginal
8	A. Rode, A. Hildebrandt, B. Schmidt	IFORM	GHM	marginal
9 DS	E. Vanem, A. B. Huseby	DSCM	GHM	marginal
9 DS s.	E. Vanem, A. B. Huseby	DSCM, smoothed	GHM	marginal
9 IFORM	E. Vanem, A. B. Huseby	IFORM	GHM	marginal

2. The benchmark and its counting procedure

2.1. The exercise

Exercise 1 of the benchmark (Haselsteiner et al., 2021) concerns six datasets. Datasets A, B and C are hourly sea states (H_s, T_z) from three NDBC buoys; D, E and F are hourly wind–wave states (u_{10}, H_s) at three points of the coastDat hindcast distributed by the World Data Center for Climate. Participants received the "provided" part of each — ten years for A–C, twenty-five for D–F — and the organizers kept a "retained" part of the same length. Contours were requested for return periods of 1 and 20 years on A–C and 1 and 50 years on D–F. Nine groups contributed; one of them submitted three sets of contours (direct sampling, direct sampling smoothed, and IFORM), so that 11 contour sets are compared, listed in Table 1 with the organizers' classification of each as a marginal-exceedance or a total-exceedance construction (Mackay and Haselsteiner, 2021). 8 sets hold twelve files (six datasets at two return periods) and 3 hold eighteen (the third set also supplied 50-year contours for A–C and 20-year ones for D–F); 150 files in all, which are the objects of this paper.

2.2. The datasets

Table 2 gives the six datasets as the ledger read them. The provided and retained parts of A–C together hold 175,320 hourly states each, which

Table 2: The six datasets: the variables, the number of hourly states in the provided and the retained part with the calendar years each spans, their total, and the highest significant wave height observed in the full dataset with its hour. Every value is read from the pinned files; the maximum is a literal compared as a literal.

set	variables	provided	years	retained	years	full	max H_s (m)
A	H_s, T_z	82,805	1996–2005	92,515	2006–2017	175,320	11.7976 (2010-02-26 05:00)
B	H_s, T_z	83,917	1996–2005	91,403	2006–2017	175,320	9.7975 (1999-09-15 07:00)
C	H_s, T_z	81,749	1996–2005	93,571	2006–2018	175,320	11.2460 (2002-10-02 21:00)
D	u_{10}, H_s	219,144	1965–1989	219,144	1990–2014	438,288	10.8044 (1993-01-24 11:00)
E	u_{10}, H_s	219,144	1965–1989	219,144	1990–2014	438,288	13.5713 (1990-12-12 12:00)
F	u_{10}, H_s	219,144	1965–1989	219,144	1990–2014	438,288	16.5892 (2005-01-12 10:00)

is 20×8766 hours to the hour; those of D–F hold 438,288 each; 1,840,824 hourly states in all. The retained parts of the buoys hold their ten years of hours over a longer span of calendar years, because a buoy record has gaps; the counts of hours are what the expected numbers use. The highest sea state of dataset A, 11.7976 m on 2010-02-26 05:00, lies in the retained years, 4.70 m above anything in the provided ten.

2.3. The submitted files

A submission is a semicolon-separated text file: a header naming the two columns, then one vertex per line as decimal literals. The column order differs between authors — some put wave height first, some second — and the organizers' scripts carry a hand-coded list of who did which (participants 8 and 9 read as period-then-height on A–C; participants 1, 2, 3, 5, 6 and 8 read as height-then-wind on D–F). The present reading takes the order from each file's header instead, which one file spells "heigth"; the two readings agree on all 150 files. Files hold between 24 and 2,052 vertices. 49 of them do not repeat their first vertex at the end, so the polygon that was counted contains an edge, from the last listed vertex back to the first, that the author did not list; the organizers' test closes the path implicitly (Section 2.4), and so does this paper, because that is the polygon the count was made on.

2.4. The organizers' count

The two scripts that produced the benchmark's Tables 5 and 6, `create_points_outside_table_abc.py` and `create_points_outside_table_def.py`, are held in the repository with the version of `viroconcom` (Haselsteiner et al., 2019) they call. For each contour set and dataset the script concatenates the provided and retained hours, reads the contour file in the column order it assumes for that participant, and calls `points_outside`, which builds a matplotlib `Path` from the vertices and asks `contains_points` for

every hour (Hunter, 2007): a crossing-number (even-odd) test in double precision on the polygon closed implicitly from its last vertex to its first, with no defined answer for a point on the boundary. An observation is outside when the test says it is not inside. A second count keeps only the observations outside with $H_s > 1$ m (A–C) or with $u_{10} > 1$ m/s and $H_s > 1$ m (D–F), the organizers' severity threshold. For the long return period the tables print the mean over the three datasets and the three per-dataset integers, for both counts; for the one-year contours they print the two three-dataset means to one decimal; and beside each the expected number of the benchmark's Table 1 (Section 3.5). That is $11 \times (3 + 3 + 1 + 1) = 88$ numbers per table, 176 in all.

2.5. What is compared

The numbers re-decided here are those of the preprint held in the benchmark's repository at the pinned commit (`2021-01-19_EC_Benchmark_Joint_Paper_WithFrontPages.pdf`, 6,899,667 bytes, SHA-256 `aff4e7397448…`), transcribed digit for digit into the record; the published article's tables were not read, and whether they differ from the preprint's is an open point for the review. The contour files, the organizers' scripts and the twelve datasets are pinned at the same commit, `a1561fe7`, by their SHA-256 digests (167 files, fetched 2026-09-12).

3. The exact re-decision

3.1. Literals, not doubles

Every number in the benchmark is a decimal literal: an hourly state "0.2845; 4.7252", a vertex "4.283446918632201; 7.469377172830912". Each is read as the rational its digits denote and kept as that string beside its double image. The fast path of every predicate runs on the doubles; the exact path, entered only where the fast path cannot decide, scales each literal to an integer at a fixed decimal scale (sixty-four places; the benchmark's longest literal has seventeen) and computes in arbitrary-precision integers, where every sign is exact. A literal that would need more places than the scale carries is refused rather than rounded; so is a token that is not a decimal literal, such as `nan`.

3.2. The polygon a file denotes

A file's vertices, in the frame (u, h) with u the second variable and h the wave height, define a polygon closed from the last vertex to the first. A

file that repeats its first vertex at the end denotes the same polygon and is recorded as explicitly closed; a vertex repeated consecutively is a zero-length edge, removed and counted; fewer than three distinct vertices is not a polygon and is refused. For each polygon the record holds, exactly: whether any two non-adjacent edges cross properly (each segment's endpoints on strictly opposite sides of the other), whether any touch, whether any adjacent pair folds back along itself, the first crossing pairs by edge index, whether the implicit closing edge is among the crossing edges, its length and whether it is the longest edge, the signed area, and the maximum of h along the contour, which is attained at a vertex because h is linear on an edge. A polygon with no crossing, touch or fold is *simple*; the others are reported as such and counted all the same, because the organizers' test counted them.

3.3. Classification: inside, outside, on

A point is classified by the even-odd crossing number of a horizontal ray (Shimrat, 1962; Hormann and Agathos, 2001), in the half-open convention of the organizers' library: an edge counts when exactly one of its endpoints lies strictly above the ray. Every predicate is a sign — of the 2×2 determinant that orients a point against an edge, or of the difference of two literals. Each is evaluated first in double precision and trusted only when its magnitude clears a forward-error bound proved for the inputs at hand (Higham, 2002; Fortune and Van Wyk, 1996): with every coordinate of magnitude at most M and $u = 2^{-53}$, each difference of two literals carries an error of at most $4uM$, each product at most $20uM^2$, and the determinant at most $48uM^2$, which is 8.7×10^{-11} at the $M = 128$ that covers the benchmark's coordinates; a computed determinant larger than 10^{-10} in magnitude therefore has the right sign, and one smaller is re-decided exactly. A point found on an edge or at a vertex is ON, a third answer the float test does not have. The nonzero-winding count is kept beside the even-odd answer; the two can differ only where a contour crosses itself, and the record says whether they do. The filter is the ordinary static filter of exact geometric computation (Fortune and Van Wyk, 1996; Shewchuk, 1997; Yap, 1997); what is particular here is only that it is written against the literals the files contain rather than against doubles, so that a vertex printed to three decimals and an observation printed to four can be found to coincide.

3.4. The counting procedure, reproduced

For each contour the provided, the retained and the full (concatenated) datasets are classified, and the counts outside, inside, on, outside above the severity threshold, and outside by the winding rule are recorded with the

hours of the first few ON points. The printed tables are then read against the full-dataset counts by one rule: a printed integer *agrees* when it lies in $[n_{\text{out}},\, n_{\text{out}} + n_{\text{on}}]$, the boundary points being the ones a float test may put on either side, and is *exact* when it equals n_{out}; a printed one-decimal mean agrees when it lies within the rounding of the three-dataset mean with the boundary points on either side, and is exact when it is the rounding of the mean of the exact counts. Nothing else is tolerated: a printed number that agrees under neither reading is recorded as *not the count of the file on record*, with the exact counts beside it.

3.5. The expected numbers, enclosed

The benchmark prints beside every count the number expected for the contour's definition, $E = n\,\alpha_t$, with $\alpha = 1/(T \times 365.25 \times 24)$ the hourly exceedance probability of a T-year contour and α_t the probability that the contour is exceeded anywhere: α itself for a total-exceedance construction, and for IFORM $\alpha_t = 1-\chi^2_2(\beta^2) = \exp(-\beta^2/2)$ with $\beta = \Phi^{-1}(1-\alpha)$, because the IFORM contour is the image of the circle of radius β in standard normal space and the squared norm of a standard bivariate normal is χ^2_2 (Haselsteiner et al., 2021; Mackay and Haselsteiner, 2021). The total-exceedance value is an exact rational. The IFORM value needs Φ^{-1}, which is enclosed: erfc on $x \geq 1$ by Laplace's continued fraction (Abramowitz and Stegun, 1964, §7.1.14), whose elements are positive so that consecutive convergents bracket the value, both evaluated in outward-rounded interval arithmetic (Moore et al., 2009; Rump, 2010); β by bisection on $\frac{1}{2}\operatorname{erfc}(\beta/\sqrt{2}) - \alpha$ that tightens only where the sign is certain; and $\exp(-\beta^2/2)$ and the product with n as interval arithmetic. The result is an interval containing the expected number, and the printed value is read as a rounding of it or not.

3.6. The record and its controls

The decision is one JavaScript module with no dependencies, so that the page which lets a reader classify any sea state in a browser runs the same bytes as the ledger. A battery of 218 checks runs before the ledger is accepted, 14 of them red controls that must fire: among them a determinant, found by random search, whose double-precision value is $+3.6 \times 10^{-15}$ and whose exact sign is negative, which the filter must send to exact arithmetic; a bow-tie that must be reported as one crossing; a square traced twice, on which even-odd and winding must part company; a fold; a `nan` token; the two columns swapped, which must move a planted point across the contour; and a point 10^{-20} outside an edge, which must not be ON. The ledger runner re-hashes the 167 pinned files, fetching the datasets from the pinned commit

when absent, and re-decides the 150 contours against the 1,840,824 hours in 5.9 s on a laptop: 637,402,203 predicates, of which 237,173 (372 per million) fell below the filter's bound and were decided in integer arithmetic. The shipped ledger re-derives identically from the bytes on every check.

4. Results

4.1. The printed counts

Tables 3 and 4 give, for every contour set and dataset, the exact count outside the long-return-period polygon with the printed count where it differs, the exact count above the severity threshold, and the exact one-year means. Fig. 1 reads all 176 printed numbers against the exact ones. 173 are the exact count to the integer, or the exact mean to the digit printed. One more agrees within the boundary points: the one-year wind–wave mean of contribution 2 (Section 4.3). Two, the one-year sea-state row of contribution 3, agree under neither reading (Section 4.2). The organizers' arithmetic is, in other words, reproduced wherever the file they counted is the file on record; what the exact re-decision adds in those 173 cells is not a different number but the standing of the number as a decision, with the polygon's digest, the rule and the boundary points recorded beside it.

4.2. Two numbers traced to a rewritten file

The one-year sea-state contours of contribution 3 give 147, 106 and 59 observations outside on A, B and C, a mean of 104.0 against a printed 82.3; above the threshold 26, 100 and 39, a mean of 55.0 against a printed 48.3. The repository's history, read from the GitHub commits API on 2026-09-12, says why. The three files were added on 2020-08-11 (commit `7eea939a`) with 360 rows each, and rewritten on 2020-09-25 (commit `07f0ecc3`, "Remove pairs where on coordinate is nan from contribution 3 as these are not valid coordinates"), which removed 53, 41 and 55 rows in which one coordinate was `nan`, leaving 307, 319 and 305. The 20-year files of the same contribution have a single commit, and their counts, 68, 28 and 33, are the printed ones to the integer, as are their counts above the threshold. A vertex with a coordinate that is not a number has no place in a crossing test, and what a floating-point path returns for a polyline that contains one is a property of the library and its version, not of any polygon; the printed row therefore corresponds to no polygon that can now be constructed from the repository, and the exact decision refuses to reproduce it rather than reproduce it approximately. The verdict recorded is *not the count of the file on record*: a statement of provenance, not of error in the organizers'

Table 3: Sea-state datasets A, B, C, the 20-year contours on the full 175,320 hours (sets as in Table 1): the exact one-year means of observations outside, and outside with $H_s > 1$ m (the benchmark prints only these means; the printed value follows in parentheses where it differs); the exact count outside the 20-year polygon for A, B and C (printed value in parentheses where it differs; † marks a polygon that is not simple); and the same counts above the threshold $H_s > 1$ m. T, total-exceedance construction; M, marginal-exceedance. The counts on the retained years alone are in Appendix B.

	1-year, mean of A, B, C		20-year, outside			20-year, $H_s > 1$ m		
set	outside	$H_s > 1$ m	A	B	C	A	B	C
1 (T)	256.3	241.0	153	127	62	117	119	60
2 (M)	406.7	389.3	437†	236†	187	401	228	185
3 (T)	104.0 (82.3)	55.0 (48.3)	68	28	33	14	26	30
4 (T)	75.0	63.0	1†	32	8	1	30	4
5 (M)	18,295.3	495.3	8,503†	12,573	29,216	47	24	3
6 (M)	154.0	115.3	27	14	9	20	12	8
7 (M)	281.3	63.0	79	4†	23	21	0	2
8 (M)	762.0	638.0	189	368	409	93	339	408
9 DS (M)	22,127.3	130.7	22,504†	22,933†	20,462†	4	43	18
9 DS s. (M)	12,031.7	126.7	5,096†	7,716†	5,376†	4	42	17
9 IFORM (M)	22,207.7	233.7	22,514	22,990	20,480	8	100	35

Table 4: Wind–wave datasets D, E, F, the 50-year contours on the full 438,288 hours, as Table 3; the threshold is $u_{10} > 1$ m/s and $H_s > 1$ m.

	1-year, mean of D, E, F		50-year, outside			50-year, threshold		
set	outside	threshold	D	E	F	D	E	F
1 (T)	424.3	268.3	154	86	71	107	63	11
2 (M)	1,185.7 (1,186.3)	744.0	514†	720	355†	304	388	236
3 (T)	524.7	341.3	154	82	72	106	62	12
4 (T)	88.0	88.0	3	8	0	3	8	0
5 (M)	2,235.0	860.0	174	112	182	7	18	19
6 (M)	270.7	247.7	3	17	5	3	17	5
7 (M)	371.3	258.0	3	16	7	2	16	2
8 (M)	1,729.7	1,442.0	509	110	167	449	108	159
9 DS (M)	18,191.7	1,941.3	17,460	17,226†	11,217	146	122	41
9 DS s. (M)	12,819.3	1,952.3	5,092	6,782	5,440	146	122	41
9 IFORM (M)	16,267.3	686.7	17,415	17,184	11,202	123	107	29

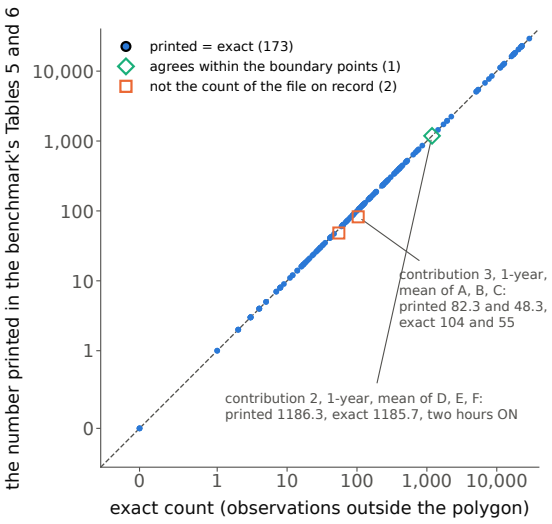


Figure 1: The 176 numbers printed in the benchmark's Tables 5 and 6 against the exact counts (both axes symmetric-logarithmic). 173 lie on the diagonal; the one that agrees within the two boundary points and the two that are not the count of the file on record are marked.

arithmetic, which on the files that were not rewritten is reproduced to the integer in every cell.

4.3. Two observations exactly on a contour

Two hourly observations lie exactly on a submitted contour, both on contribution 2's one-year wind–wave contours: in dataset E the state of 1989-06-17 20:00 ($H_s = 0.1240$ m, $u_{10} = 2.4972$ m/s) and in dataset F the state of 1986-06-27 20:00 ($H_s = 0.2280$ m, $u_{10} = 1.9572$ m/s), both in the provided years. The vertices of those contours are given to 3 decimals and the observations to four, and they coincide. The exact counts outside are 1,252, 1,477 and 828 on D, E and F, a mean of 1185.7; the printed mean is 1186.3, which is the mean once the two boundary points are counted outside (1186.3). That is what the float test does: run today with matplotlib 3.11.2 on the same files it returns 1,478 and 829 outside on E and F, the boundary point on the outer side in both. The record does not pick a side; it says ON, and reads the printed mean as consistent with either. One more digit on either observation and it would be inside or outside; the battery plants exactly that case.

4.4. Polygons that cross themselves, and the edge nobody listed

A point-in-polygon test presumes a polygon. 30 of the 150 files describe a closed polyline whose edges cross each other properly (none merely touches or folds): 10 of contribution 2's 12 direct-sampling contours, which list 24 vertices each with 44 consecutive duplicates among them; 9 of contribution 9's 18 direct-sampling contours and 6 of its 18 smoothed ones, with between 1 and 26 crossings each; 2 of contribution 5's 12 declustered direct-IFORM contours; and, with one crossing each, 2 of contribution 7's IFORM contours and the highest-density contour of contribution 4 for dataset A at 20 years. Appendix A lists all thirty. Fig. 2c draws the direct-sampling 20-year contour for A, vertex to vertex in the order the file lists them: a three-cornered loop of 60 distinct vertices whose edges cross 21 times, the crossings clustered at the corners where successive sampling directions return nearly the same boundary point and the listed order zigzags at the scale of centimetres. The highest-density case is of another kind (Fig. 2d): its 1,160 listed vertices trace a simple staircase along the cells of a grid, and the one crossing is the implicit closing edge — 0.112 long in the plane's units, the longest edge of the polygon — cutting across the curve's own tail, which the author's software had not joined. In 11 of the 30 the closing edge is one of the crossing edges.

Under the even-odd rule a self-crossing polygon still has an inside, but it is not the region a reader sees when the curve is drawn: the loops between crossings alternate in and out, and the nonzero-winding rule would fill them. Both rules were applied to every observation on every contour, and no observation is classified differently by the two, so the printed counts do not depend on the choice of rule. What depends on it is what the number is a count of. 49 files are closed by an edge the author did not list (contribution 2 (11 of 12), contribution 3 (12 of 12), contribution 4 (12 of 12), contribution 5 (12 of 12) and contribution 7 (2 of 12)); for most it is short, and for some contours of contributions 4 and 5 it is the longest edge of the polygon. These are properties of the files as submitted, decided exactly and recorded as such; whether they are properties of the constructions that produced the files is not decided here, and a reader of the benchmark who wants to know what a count means for a given file now has the file's diagnostics beside it.

4.5. The expected numbers, certified

Table 5 gives the enclosures. On the sea-state datasets the total-exceedance expectation is exactly 20 for a one-year contour and exactly 1 for twenty years, because the full datasets hold 175,320 = 20 × 8766 hours; on the

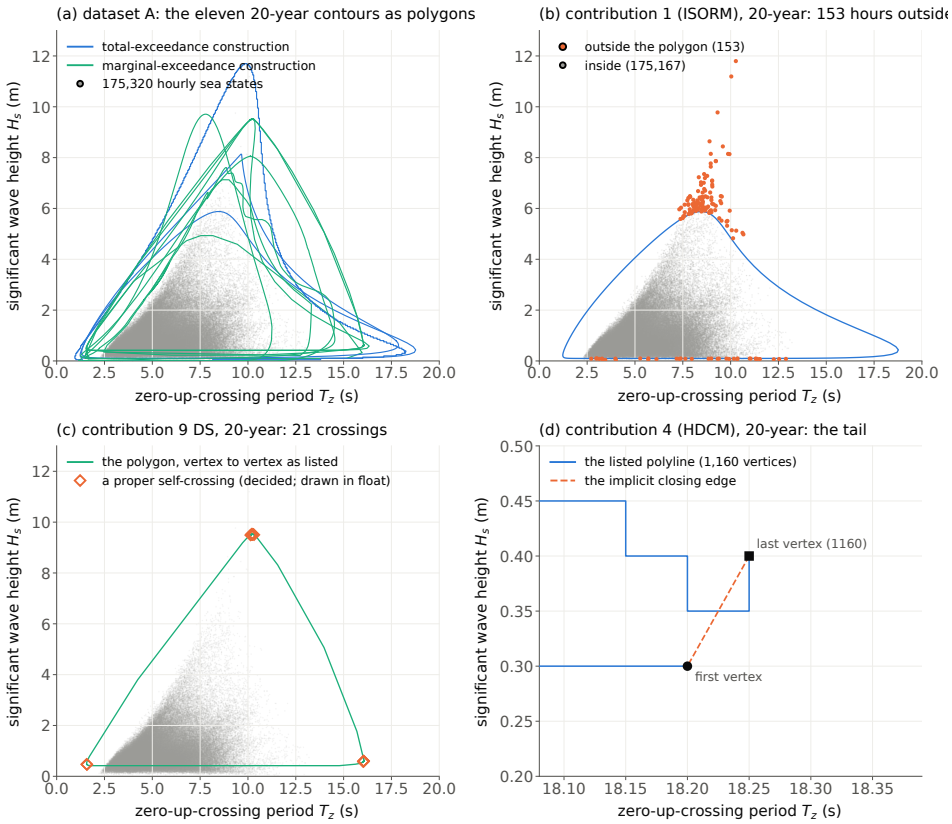


Figure 2: Dataset A. (a) The eleven 20-year contours as the polygons their files denote, each closed from its last vertex to its first, over the 175,320 hourly sea states. (b) One of them, contribution 1's ISORM contour, with the 153 hours decided outside it marked; the count is the ledger's, and the drawing refuses to build if matplotlib's own test on the same polygon does not reproduce it. (c) Contribution 9's direct-sampling contour drawn vertex to vertex in file order, with its 21 proper self-crossings (the count is decided; the positions are drawn in floating point). (d) The tail of contribution 4's highest-density contour: the listed polyline ends at the square and the implicit closing edge back to the circle, the longest edge of the polygon, crosses the staircase.

Table 5: The benchmark's Table 1 certified: for n hourly states and a T-year contour, the hourly exceedance probability α as an exact rational, the total-exceedance expectation $n\alpha$ (exact) against the printed value, and the IFORM expectation $n\exp(-\beta^2/2)$ with $\beta = \Phi^{-1}(1-\alpha)$ as enclosures against the printed value. Where an enclosure's two ends agree to the digits shown, one number is printed.

datasets	T (yr)	n	α	$n\alpha$	printed	β	IFORM E	printed
A, B, C	1	175,320	$\frac{1}{8766}$	20	20	3.6856114265	196.862	197
A, B, C	20	175,320	$\frac{1}{175320}$	1	1	4.3886106004	11.524	11.5
D, E, F	1	438,288	$\frac{1}{8766}$	$\frac{73048}{1461} = 49.9986$	50	3.6856114265	492.141	492
D, E, F	50	438,288	$\frac{1}{438300}$	$\frac{36524}{36525} = 0.9999$	1	4.5839339344	11.994	12

wind–wave datasets it is $73048/1461 = 49.9986$ and $36524/36525 = 0.9999$, which the benchmark prints as 50 and 1. For IFORM, $\beta = 3.6856114265$ for the one-year contour and 4.3886106004 for twenty years, both ends of the enclosure agreeing to the digits shown, giving $E = 196.862$ and 11.524 — the benchmark's 197 and 11.5 — and on the wind–wave datasets 492.141 and 11.994, its 492 and 12. The out-of-sample expectation follows from the same formula: a long-return-period total-exceedance contour fitted on the provided years should be exceeded $n_{\text{retained}}\,\alpha$ times on the retained years, 0.5276, 0.5213 and 0.5337 on A, B and C and $\frac{18262}{36525} = 0.4999$ on each of D, E and F; an IFORM contour between 6.00–6.15 times.

4.6. Numbers the benchmark did not print

Appendix B gives the count on the retained years alone — the years the contributors did not see — for every long-return-period contour. The highest-density contours of contribution 4 are exceeded 1, 2, 2, 1, 6 and 0 times on A–F, 12 in all against an expectation of 3.08 summed over the six; the ISORM contours of contribution 1, built for the same exceedance, 107, 36, 33, 104, 48 and 65 times; the inverse-directional contours of contribution 3, 57, 4, 14, 104, 45 and 66. The benchmark's own caution applies with more force here than to its full-data counts: hourly sea states are serially correlated, a storm that crosses a contour is counted at every hour it stays outside, and these are counts, not tests (Haselsteiner et al., 2021). The second number is the maximum along each contour (Fig. 3): in dataset A the highest observed state, 11.7976 m on 2010-02-26 05:00, sits above the maximum of every one of the eleven 20-year contours, the highest of which, contribution 4 (HDCM), reaches 11.70 m; that contour has exactly 1 observation outside it on the full dataset, which is that hour. The benchmark says as much from a figure; here each maximum is a literal read from the file. Over the six datasets, 27 of the 150 contours reach above the highest

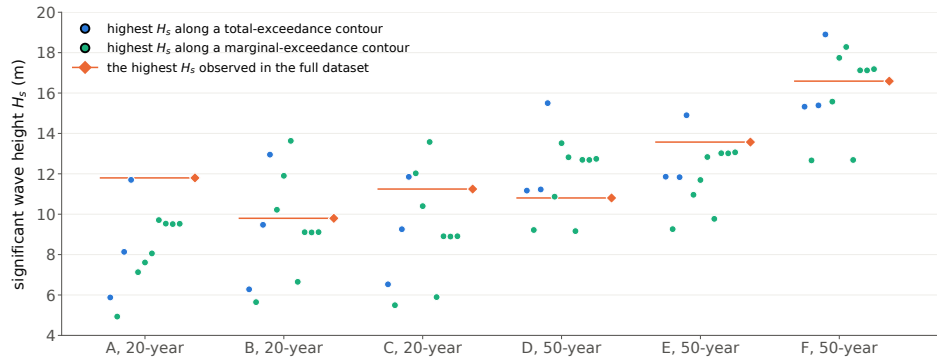


Figure 3: For each dataset, the highest significant wave height along each of the eleven long-return-period contours (a literal read from the file: a linear function on a polygon peaks at a vertex) against the highest wave height observed in the full dataset.

observed wave; among the long-return-period contours, none on A and between 1 and 9 of the eleven on each of the other five (A 0, B 4, C 3, D 9, E 1, F 6).

4.7. The printed marginals

The benchmark's appendix prints the H_s marginal each contributing team fitted to the provided ten years of A, B and C. Those printed digits are decided in the companion paper (Toledo, 2026) by a different instrument — a maximum-likelihood fit certified as the likelihood's one maximum in a box, the printed parameters read as the box their digits allow — and only the result is recalled here: 15 printed rows from 5 contributions can be read as a family and decided on the hours they were fitted to; on buoy A they put the hourly 100-year level anywhere between 5.25 and 14.53 m, where the certified maximum-likelihood fits of the Weibull, the lognormal and the exponentiated Weibull on the same 82,805 hours give 5.25, 12.11 and 15.90 m; one contribution's two-parameter Weibull is the certified fit to its last printed digit on A and B and, on C, the certified fit of the period rather than of the wave height; one translated Weibull leaves 30,740 observed hours of A, B and C below its printed location. None of this is visible in the printed digits and none of it bears on the counts above, which take each contour as submitted.

5. Discussion

What a certificate adds to a benchmark of this kind. For 173 of 176 printed numbers the exact re-decision returns the number the organizers printed, and a reader may ask what was gained. Three things. The number is now a decision: for each contour the record holds the file's digest, the polygon it denotes with its diagnostics, the rule and its boundary convention, the counts under both rules, and the hours of the boundary points, and anyone can re-derive the integer from those bytes in any exact arithmetic. Refusal is a result: a file that does not denote a simple polygon is reported, with its crossings, rather than counted over, and a file that no longer denotes the polygon that was counted is traced rather than reproduced approximately. And the boundary has a name: two of the 1,840,824 classifications are ON, which no float test can say, and the one printed mean they touch is read as consistent with either side rather than as right or wrong. None of this changes which construction looks better in the tables. It changes what the tables are: counts a reader must take on trust become counts a reader can check.

What the next exercise could require. Two requirements would make a future benchmark of contours decidable by construction, at no cost to the participants. The first concerns the files: a contour should be submitted as an explicit polygon — closed, with no vertex repeated consecutively, no coordinate that is not a number, a header naming the columns — and the organizers' reader should refuse anything else, or report it, before counting; whether a simple polygon is required or a self-crossing one is accepted with its crossings declared is a choice the exercise should make in writing, since 30 of 150 submissions here would have been affected. The second concerns the count: the rule for "inside" (even-odd or winding), the convention for a point on the boundary (counted inside, outside, or reported), the concatenation of the data, the thresholds, and the arithmetic should be declared in the call for contributions, and the count itself published as an integer per contour per dataset with the file's digest beside it, rather than as a three-dataset mean. An exact point-in-polygon decider is a few hundred lines in any language and runs over the whole exercise in seconds; the organizers' float count reproduced the exact one in every cell where the file was the file on record, so the change is not to the arithmetic but to what is written down. A third, cheaper requirement follows from Section 3.5: the expected number beside each count should be printed with its enclosure or, where it is a rational, exactly, and an out-of-sample column on the retained years should be printed beside the full-data one.

What is not claimed. No contour is called right or wrong. Which construction an offshore designer should use is a modelling question that this paper does not touch; a count of exceedances is the benchmark's own metric, taken here on its own terms, and the benchmark itself notes the serial correlation that makes such counts rough (Haselsteiner et al., 2021). The two numbers that are not the count of the file on record are not an error in the organizers' arithmetic; the file they were computed from is not the file the repository holds. A self-crossing polygon is a property of a file as submitted, not necessarily of the method that produced the boundary it lists, and nothing here measures how far the region counted differs from the region intended. The enclosures of Section 3.5 certify the benchmark's formula, not the independence it assumes. The numbers compared are the preprint's; the published article was not read. The datasets are NDBC's and the World Data Center for Climate's and are not redistributed; the contours are held verbatim as the published record, for verification only. And in the regulatory sense nothing here certifies anything: a design basis is accepted by a classification society under national rules (DNV, 2021; ISO, 2015), and the word is used in this paper for machine-checkable statements only.

Cost. The whole exercise is re-decided in 5.9 s on a laptop: 6.37×10^8 predicates, of which 237,173 fell below the filter's bound. The exact path costs nothing a benchmark would notice; the cost of the method is in writing the filter's bound down and in planting the controls that must fire.

6. Conclusions

The 150 contours submitted to Exercise 1 of the environmental-contour benchmark of Haselsteiner et al. (2021) were read as the exact polygons their files denote and every one of the 1,840,824 hourly sea states they were tested against was classified inside, outside or on each, with every sign decided. The organizers' counting procedure, applied as their script defines it, reproduces 173 of their 176 printed numbers to the integer; one more within the two observations that lie exactly on a contour; and two, the one-year sea-state row of one contribution, are traced through the repository's history to three files rewritten after they were counted. 30 of the 150 polygons cross themselves and 49 are closed by an edge nobody listed, properties of the files that bear on what each count is a count of and that no float count reports; no observation changes side between the two rules for "inside". The benchmark's expected numbers are the roundings of certified enclosures, two of them exact. What is proposed to the metocean community is not a verdict

on any contour method but a different object to accompany a benchmark's number: an explicit polygon, a declared rule, and a record in which the count is a decision and a file's defects are a result.

Declaration of competing interest

The author declares no competing financial or personal interests. The benchmark's organizers and contributors have not been consulted on this draft.

Data availability

The records, the decider and the figure and table sources are public in the cert-machine repository (`https://github.com/carlostoledo1891/cert-machine`, commit 9349d8bad505): the ledger `certs/ecbench-ledger.json` (generated 2026-09-12), the pinned corpus `corpus/ec-benchmark` (the 150 contour files verbatim, the organizers' scripts, the preprint's tables transcribed in `claims.json`, the pin of the twelve datasets, which are fetched from commit `a1561fe7` of `https://github.com/ec-benchmark-organizers/ec-benchmark` and verified by digest), the decider `instruments/ecbench` with its battery (MIT), the ledger runner `tools/run-ecbench-ledger.js`, and the manuscript's number and figure builders `tools/paper-numbers/ec-benchmark.js` and `ec-benchmark-figs.py`. The report with every sentence gated on the ledger is at `https://carlostoledo.co/reports/ec-benchmark.html`, and the instrument in which any sea state can be classified against any submitted contour in the browser, by the same bytes, at `https://carlostoledo.co/instruments/contours`. The companion decision of the printed marginals is in `certs/hseva-ledger.json` (2026-10-09).

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used large language models (Claude, Anthropic) to draft code, exposition and this manuscript in a certified-verification setting: constructions and prose were proposed with the help of the models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for a record to be accepted. After using the tool the author reviewed and edited the content and takes full responsibility for the content of the publication. The records, not the models, are the evidence offered; they are public and re-runnable.

Funding

This work received no external funding.

Appendix A. The thirty polygons that are not simple

Table A.6: Every submitted contour whose edges cross properly: the listed vertices, the distinct vertices after consecutive duplicates are removed, the duplicates, whether the file repeats its first vertex, the number of proper crossings, whether the implicit closing edge is a crossing edge, and the closing edge's length in the plane's units (∗: the longest edge of the polygon).

set	dataset / yr	listed	distinct	dupl.	closed	crossings	closing edge crosses	closing edge
2 DSCM	A / 1	24	22	2	no	3	yes	0.124
2 DSCM	A / 20	24	15	9	no	1	yes	0.253
2 DSCM	B / 1	24	22	2	no	3	no	0.045
2 DSCM	B / 20	24	18	5	yes	2	no	0.552
2 DSCM	C / 1	24	21	3	no	2	no	0.043
2 DSCM	D / 1	24	22	2	no	3	no	0.004
2 DSCM	D / 50	24	19	5	no	1	no	0.060
2 DSCM	E / 1	24	22	2	no	2	no	0.044
2 DSCM	F / 1	24	24	0	no	1	no	0.693
2 DSCM	F / 50	24	20	4	no	3	yes	1.170
4 HDCM	A / 20	1,160	1,160	0	no	1	yes	0.112*
5 DIFORM	A / 1	52	52	0	no	9	yes	1.446
5 DIFORM	A / 20	66	66	0	no	7	no	1.667
7 IFORM	B / 20	1,374	1,374	0	no	1	no	0.000
7 IFORM	E / 1	790	790	0	no	1	no	0.000
9 DS DSCM	A / 1	61	60	0	yes	1	yes	0.072
9 DS DSCM	A / 20	61	60	0	yes	21	yes	0.063
9 DS DSCM	A / 50	61	60	0	yes	26	yes	0.172
9 DS DSCM	B / 20	61	60	0	yes	7	no	0.011
9 DS DSCM	B / 50	61	60	0	yes	19	no	0.192
9 DS DSCM	C / 20	61	60	0	yes	4	no	0.009
9 DS DSCM	C / 50	61	60	0	yes	12	yes	0.129
9 DS DSCM	E / 1	61	60	0	yes	1	no	0.375
9 DS DSCM	E / 50	61	60	0	yes	1	yes	0.817
9 DS s. DSCM, smoothed	A / 20	121	120	0	yes	12	no	0.004
9 DS s. DSCM, smoothed	A / 50	121	120	0	yes	17	no	0.062
9 DS s. DSCM, smoothed	B / 20	121	120	0	yes	7	yes	0.003
9 DS s. DSCM, smoothed	B / 50	121	120	0	yes	8	no	0.058
9 DS s. DSCM, smoothed	C / 20	121	120	0	yes	1	no	0.008
9 DS s. DSCM, smoothed	C / 50	121	120	0	yes	2	no	0.041

Appendix B. The retained years alone

Table B.7: Observations outside each long-return-period contour on the retained years alone (20-year contours on A–C, 50-year on D–F), with the expected number for a total-exceedance and for an IFORM contour (Section 3.5). T, total-exceedance construction; M, marginal-exceedance.

set	A	B	C	D	E	F
1 ISORM (T)	107	36	33	104	48	65
2 DSCM (M)	275	84	86	276	346	206
3 IDSCM (T)	57	4	14	104	45	66
4 HDCM (T)	1	2	2	1	6	0
5 DIFORM (M)	4,774	6,189	15,771	78	48	132
6 IFORM (M)	27	9	3	0	14	3
7 IFORM (M)	62	3	11	0	11	4
8 IFORM (M)	93	101	183	263	59	91
9 DS DSCM (M)	12,078	11,830	11,045	7,586	7,502	5,310
9 DS s. DSCM, smoothed (M)	2,976	3,667	3,086	1,954	2,846	2,577
9 IFORM IFORM (M)	12,085	11,844	11,049	7,562	7,484	5,306
expected, total exceedance	0.5276	0.5213	0.5337	0.4999	0.4999	0.4999
expected, IFORM	6.081	6.008	6.150	5.997	5.997	5.997

References

Abramowitz, M., Stegun, I.A., 1964. Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables. National Bureau of Standards, Applied Mathematics Series 55, Washington.

Chai, W., Leira, B.J., 2018. Environmental contours based on inverse SORM. Marine Structures 60, 34–51.

DNV, 2021. DNV-RP-C205: Environmental conditions and environmental loads. Recommended practice.

Fortune, S., Van Wyk, C.J., 1996. Static analysis yields efficient exact integer arithmetic for computational geometry. ACM Transactions on Graphics 15, 223–248.

Haselsteiner, A.F., Ohlendorf, J.-H., Wosniok, W., Thoben, K.-D., 2017. Deriving environmental contours from highest density regions. Coastal Engineering 123, 42–51.

Haselsteiner, A.F., Lehmkuhl, J., Pape, T., Windmeier, K.-L., Thoben, K.-D., 2019. ViroCon: A software to compute multivariate extremes using the environmental contour method. SoftwareX 9, 95–101.

Haselsteiner, A.F., Coe, R.G., Manuel, L., Chai, W., Leira, B., Clarindo, G., Guedes Soares, C., Hannesdóttir, Á., Dimitrov, N., Sander, A., Ohlendorf, J.-H., Thoben, K.-D., de Hauteclocque, G., Mackay, E., Jonathan, P., Qiao, C., Myers, A., Rode, A., Hildebrandt, A., Schmidt, B., Vanem, E., Huseby, A.B., 2021. A benchmarking exercise for environmental contours. Ocean Engineering 236, 109504.

Higham, N.J., 2002. Accuracy and Stability of Numerical Algorithms, 2nd ed. SIAM, Philadelphia.

Hormann, K., Agathos, A., 2001. The point in polygon problem for arbitrary polygons. Computational Geometry 20, 131–144.

Hunter, J.D., 2007. Matplotlib: A 2D graphics environment. Computing in Science & Engineering 9, 90–95.

Huseby, A.B., Vanem, E., Natvig, B., 2013. A new approach to environmental contours for ocean engineering applications based on direct Monte Carlo simulations. Ocean Engineering 60, 124–135.

ISO, 2015. ISO 19901-1:2015 Petroleum and natural gas industries — Specific requirements for offshore structures — Part 1: Metocean design and operating considerations.

Jonathan, P., Ewans, K., 2013. Statistical modelling of extreme ocean environments for marine design: A review. Ocean Engineering 62, 91–109.

Kettner, L., Mehlhorn, K., Pion, S., Schirra, S., Yap, C., 2008. Classroom examples of robustness problems in geometric computations. Computational Geometry 40, 61–78.

Mackay, E., Haselsteiner, A.F., 2021. Marginal and total exceedance probabilities of environmental contours. Marine Structures 75, 102863.

Moore, R.E., Kearfott, R.B., Cloud, M.J., 2009. Introduction to Interval Analysis. SIAM, Philadelphia.

Ross, E., Astrup, O.C., Bitner-Gregersen, E., Bunn, N., Feld, G., Gouldby, B., Huseby, A., Liu, Y., Randell, D., Vanem, E., Jonathan, P., 2020. On environmental contours for marine and coastal design. Ocean Engineering 195, 106194.

Rump, S.M., 2010. Verification methods: rigorous results using floating-point arithmetic. Acta Numerica 19, 287–449.

Shewchuk, J.R., 1997. Adaptive precision floating-point arithmetic and fast robust geometric predicates. Discrete & Computational Geometry 18, 305–363.

Shimrat, M., 1962. Algorithm 112: Position of point relative to polygon. Communications of the ACM 5, 434.

Toledo, C., 2026. Certified return levels: deciding maximum-likelihood extreme-value fits of significant wave height over a public global hindcast. Companion manuscript, draft for review of 2026-10-01, in the cert-machine repository.

Winterstein, S.R., Ude, T.C., Cornell, C.A., Bjerager, P., Haver, S., 1993. Environmental parameters for extreme response: inverse FORM with omission factors. In: Proceedings of the 6th International Conference on Structural Safety and Reliability (ICOSSAR 93), Innsbruck.

Yap, C.K., 1997. Towards exact geometric computation. Computational Geometry 7, 3–23.